# The Jordan–Wigner Transformation

Lecture Notes

# Outline

# Fermionic Operators and Second Quantization

- **Fermionic modes:** Defined by creation ($a_p^\dagger$) and annihilation ($a_p$) operators for each mode $p = 0, \ldots, N-1$. These operators obey the *canonical anticommutation relations*:

  $$\{a_p, a_q\} = 0, \quad \{a_p, a_q^\dagger\} = \delta_{pq},$$

  where $\{A, B\} = AB + BA$. (Vacuum: $a_p|\text{vac}\rangle = 0$) $[\text{oai}_c itation : 0 quantumai.google](https :$
  $//quantumai.google/openfermion/tutorials/jordan_w igner_a nd_b ravyi_k itaev$
  $text =$
  $A$**Fock space basis:**$The number (occupation) basis is |n_0, \ldots, n_{N-1}\rangle =$
  $(a_0^\dagger)^{n_0}(a_1^\dagger)^{n_1} \cdots |\text{vac}\rangle, \quad n_p \in \{0, 1\}. Each a_p^\dagger$ toggles the occupancy of mode $p$ $[\text{oai}_c itation : 1 quantumai.google](https :$
  $//quantumai.google/openfermion/tutorials/jordan_w igner_a nd_b ravyi_k itaev$
  $text =, vectors)$.
- **Pauli exclusion:** Fermionic operators satisfy $a_p^2 = 0$; one cannot create two fermions in the same mode $[\text{oai}_c itation : 2 quantumai.google](https :$
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# Anticommutation and Consequences

- **Canonical Anticommutation:** $\{a_p, a_q\} = 0$ and $\{a_p, a_q^\dagger\} = \delta_{pq}$ by definition [oai$_c$*itation* : 3*quantumai.google*](*https* : //*quantumai.google*/*openfermion*/*tutorials*/*jordan*$_w$*igner*$_a$*nd*$_b$*ravyi*$_k$*itaev*
*text* = *A***Occupation operators commute:**$[\mathrm{n}_p, n_q] = 0$ with $n_p = a_p^\dagger a_p$, and each $n_p$ has eigenvalues 0 or 1 [oai$_c$*itation* : 4*quantumai.google*](*https* : //*quantumai.google*/*openfermion*/*tutorials*/*jordan*$_w$*igner*$_a$*nd*$_b$*ravyi*$_k$*itaev*
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- **Vacuum state:** There is a unique vacuum $|\mathrm{vac}\rangle$ annihilated by all $a_p$; each $a_p^\dagger$ creates a fermion in mode $p$ (raising its occupation to 1) [oai$_c$*itation* : 5*quantumai.google*](*https* : //*quantumai.google*/*openfermion*/*tutorials*/*jordan*$_w$*igner*$_a$*nd*$_b$*ravyi*$_k$*itaev*$_t$*rans*
*text* =, *eigenvector***Anticommutation implications:**$\mathrm{a}_p^2 = 0$ (Pauli exclusion) and $(a_p^\dagger)^2 = 0$. On a basis state, $a_p|n_0 \ldots 1_p \ldots n_{N-1}\rangle = (-1)^{\sum_{q<p} n_q}|n_0 \ldots 0_p \ldots n_{N-1}\rangle$ [oai$_c$*itation* : 6*quantumai.google*](*https* :

# Why Map Fermions to Qubits?

- **Quantum simulation:** We wish to simulate fermionic systems (e.g. electrons in molecules or lattice models) on a quantum computer of qubits. To do so, we need an explicit map of fermionic operators to qubit operators that preserves anticommutation [oai$_c$*itation* : 7*futureofmatter*.*com*](*https* : //*futureofmatter*.*com*/*assets*/*fermions*$_a$*nd*$_j$*ordan*$_w$*igner*.*pdf* : : *text* = *system***Spin-fermion equivalence:** *TheJordan–Wignertransformshowsspi*
  chains (qubits) can be mapped to fermions. Conversely, fermionic Hamiltonians can be expressed as qubit (spin) Hamiltonians. For example, 1D spin chains like the Ising/XY models are exactly solvable by mapping to free fermions [oai$_c$*itation* : 8*en*.*wikipedia*.*org*](*https* : //*en*.*wikipedia*.*org*/*wiki*/*Jordan*
- **Electronic structure:** Molecular electronic Hamiltonians (second-quantized fermionic operators) can be encoded into qubit Hamiltonians for quantum chemistry simulations. Each fermionic mode (spin-orbital) corresponds to one qubit under Jordan–Wigner

The Jordan–Wigner (JW) transform explicitly maps each fermionic mode to a qubit. The annihilation operator $a_p$ is mapped to a product of Pauli matrices on qubits:

$$a_p \;\mapsto\; \frac{1}{2}\,(X_p + iY_p)\,Z_0 Z_1 \cdots Z_{p-1},$$

and the creation operator $a_p^\dagger$ to its Hermitian conjugate:

$$a_p^\dagger \;\mapsto\; \frac{1}{2}\,(X_p - iY_p)\,Z_0 Z_1 \cdots Z_{p-1}.$$

That is, each fermionic mode $p$ is represented by qubit $p$, with a string of $Z$ operators on all lower-index qubits $[\mathrm{oai}_c itation : 11 quantumai.google](https :$ $//quantumai.google/openfermion/tutorials/jordan_w igner_a nd_b ravyi_k itaev_t rans$ $text =$

- This mapping ensures the correct anticommutation: the $Z$-string introduces a sign $(-1)^{\sum_{q<p} n_q}$ when swapping fermions, reproducing Fermi statistics.

Using the mapping, one can express standard fermionic operators in Pauli form:

- **Number operator:**

$$n_p = a_p^\dagger a_p = \frac{I - Z_p}{2},$$

  since $a_p^\dagger a_p$ has eigenvalue 1 if qubit $p$ is $|1\rangle$ and 0 if $|0\rangle$.

- **Fermionic parity:**

$$(-1)^{\sum_{q<p} n_q} = \prod_{k=0}^{p-1} Z_k,$$

  implemented by the string of $Z$'s in the mapping.

- **Spin operators:** The Pauli-$Z$ on qubit $j$ corresponds to $2a_j^\dagger a_j - I$ (local fermion number parity) [oai$_citation : 13futureofmatter.com](https : //futureofmatter.com/assets/fermions_and_jordan_wigner.pdf : \quad : text = operators X_j X_{j+1} + Y_j Y_{j+1} \; = \; a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j$, showing how hopping terms map to two-qubit terms.

**Inverse transform:** The Pauli operators can be expressed in terms of $a, a^\dagger$. For example,

# Example: 2-Mode Hamiltonian

Consider a two-mode fermionic Hamiltonian (e.g. tight-binding or Hubbard dimer):

$$H = \varepsilon\,(a_0^\dagger a_0 + a_1^\dagger a_1) + t\,(a_0^\dagger a_1 + a_1^\dagger a_0).$$

Under Jordan–Wigner:

$$a_0^\dagger a_0 = \frac{I - Z_0}{2}, \quad a_1^\dagger a_1 = \frac{I - Z_1}{2},$$

$$a_0^\dagger a_1 + a_1^\dagger a_0 = \frac{1}{2}(X_0 X_1 + Y_0 Y_1).$$

Thus the qubit Hamiltonian becomes

$$H_q = \frac{\varepsilon}{2}(2I - Z_0 - Z_1) \; + \; \frac{t}{2}(X_0 X_1 + Y_0 Y_1).$$

All terms are now products of Pauli operators acting on qubits 0 and 1. (This illustrates the mapping of on-site energies to $Z$ and hopping to $XX + YY$ interactions.)

# Circuit Representation of JW Operators

Fermionic operators mapped to Pauli operators can be implemented by quantum circuits:

- **Single-mode phases:** A term like $Z_p$ (from number operators) is a simple single-qubit phase-flip gate.
- **Hopping terms (XX+YY):** For example, the two-qubit interaction $e^{-i\frac{\theta}{2}(X_0X_1+Y_0Y_1)}$ can be realized by standard gate decompositions. One method: apply Hadamard gates to bring $XX$ coupling into a $ZZ$ form on a basis, perform a controlled-$Z$ rotation, and undo the basis change.
- **RYY / RXX gates:** Modern frameworks include gates like $R_{XX}(\theta) = e^{-i\theta X\otimes X}$ and $R_{YY}(\theta) = e^{-i\theta Y\otimes Y}$ $[\mathrm{oai}_c itation : 15docs.quantum.ibm.com](https : //docs.quantum.ibm.com/api/qiskit/qiskit.circuit.library.RYYGate : : text = A$**Circuit example:** $To implement e^{-i(\phi/2)X_0X_1}$, use two Hadamards and a controlled-$Z$ rotation:

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# Applications: Quantum Simulations

- **Quantum Chemistry:** Electronic structure Hamiltonians (in second quantization) map directly to qubit Hamiltonians via JW. Each molecular orbital/spin-orbital is a fermionic mode→ qubit [oai$_{c}itation : 17qiskit - community.github.io](https : //qiskit - community.github.io/qiskit - nature/tutorials/06_{q}ubit_{m}appers.html : \quad : text = The$**Spin Chains and Lattice Fermions:**$TheJWtransformisclassicforsolv$ 18$en.wikipedia.org](https : //en.wikipedia.org/wiki/Jordan$

- **Tapering via Symmetry:** In some cases, e.g. systems with fixed particle number or spin parity, the JW-mapped Hamiltonian has symmetries (global $Z$-parities) that allow qubit reduction (tapering) [oai$_{c}itation : 20qiskit - community.github.io](https : //qiskit - community.github.io/qiskit - nature/tutorials/06_{q}ubit_{m}appers.html : \quad : text = The$

# Alternative Mappings (Parity, Bravyi–Kitaev)

- **Parity mapping:** A variant of JW where parity information is stored locally on qubits. In parity mapping, occupation information is delocalized, and the qubit index encodes parity of lower modes [oai$_c$*itation* : 21*qiskit* − *community*.*github*.*io*](*https* : //*qiskit* − *community*.*github*.*io*/*qiskit* − *nature*/*tutorials*/06$_q$*ubit*$_m$*appers*.*html* : : *text* = *The***Bravyi–Kitaev (BK):***AmappingthatbalanceslocalitybetweenJWandp*
  *treestructuretoachieve*O(log $N$) Pauli-weight for parity and update operations [oai$_c$*itation* : 22*qiskit* − *community*.*github*.*io*](*https* : //*qiskit* − *community*.*github*.*io*/*qiskit* − *nature*/*tutorials*/06$_q$*ubit*$_m$*appers*.*html* : : *text* = *already*
- **Comparison:** JW is simplest (long $Z$-strings, $O(N)$ weight), parity and BK trade off Pauli string lengths differently. The best choice depends on hardware connectivity and symmetry exploitation [oai$_c$*itation* : 23*qiskit* − *community*.*github*.*io*](*https* : //*qiskit* − *community*.*github*.*io*/*qiskit* − *nature*/*tutorials*/06$_q$*ubit*$_m$*appers*.*html* : : *text* = *already*

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# References II


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